

HAT-P-20b

R-1254

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-20_241127 · created 2026-07-31T12:07:30+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2460641.7973 +/- 0.0028 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.192 +/- 0.041
Transit depth (Rp/Rs)^2	3.67 +/- 1.57 [%]
Orbital Inclination (inc)	85.74 +/- 1.24
Airmass coefficient 1 (a1)	1.012 +/- 0.031
Airmass coefficient 2 (a2)	-0.0052 +/- 0.0081
Scatter in the residuals of the lightcurve fit is	3.01 %
Best Comparison Star	#1 - [483, 229]
Optimal Aperture	3.96
Optimal Annulus	10.36
Transit Duration (day)	0.064 +/- 0.025

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.192 ± 0.041 vs 0.1284 (deviation 1.26σ)
Duration fit vs published	0.064 d vs 0.0775 d (deviation 0.44σ)
Mid-time O–C	7.5 min
Depth SNR	3.8
Residual scatter	3.01 %

Light curve

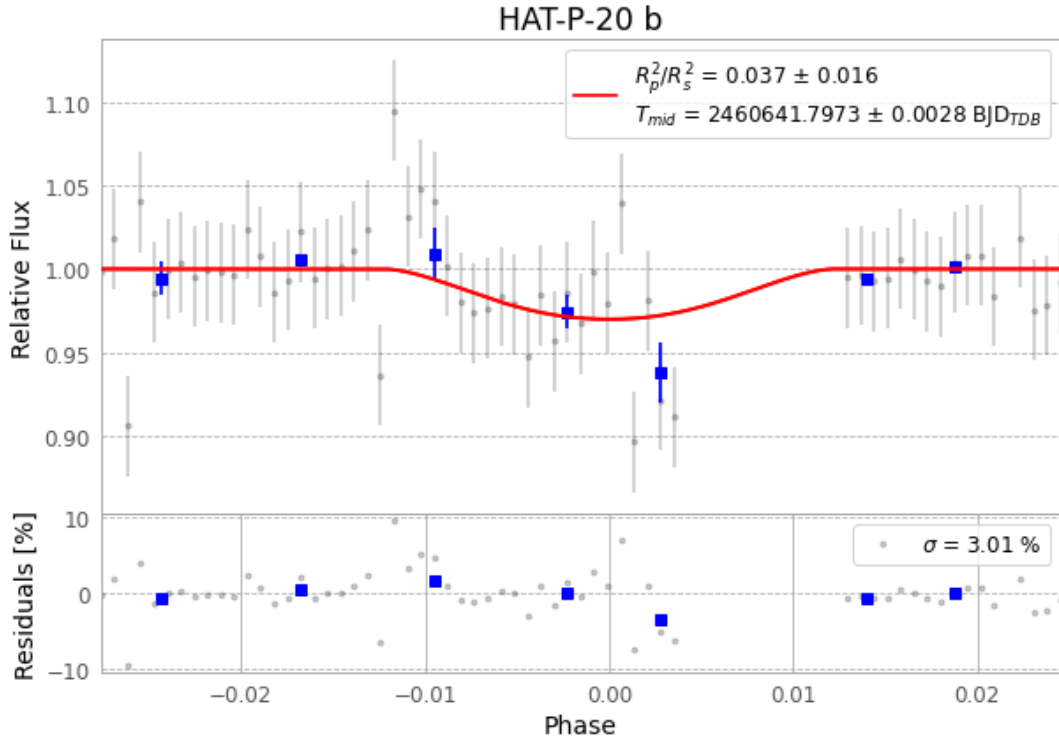


Figure 1 — FinalLightCurve_HAT-P-20 b_2024-11-26.png (SHA-256 77e4b176046c4716...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [369, 177], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 1d8cb73507ed9140...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

73 FITS frames (48 MB), HATP-20241127050616.FITS → HATP-20241127084215.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-20-241127.log (d4a324b3dd8c...), FinalLightCurve_HAT-P-20 b_2024-11-26.png (77e4b176046c...), FinalLightCurve_HAT-P-20 b_2024-11-26.csv (74f5954c3e25...)

Compute resources

Wall time 170.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 12:07Z.